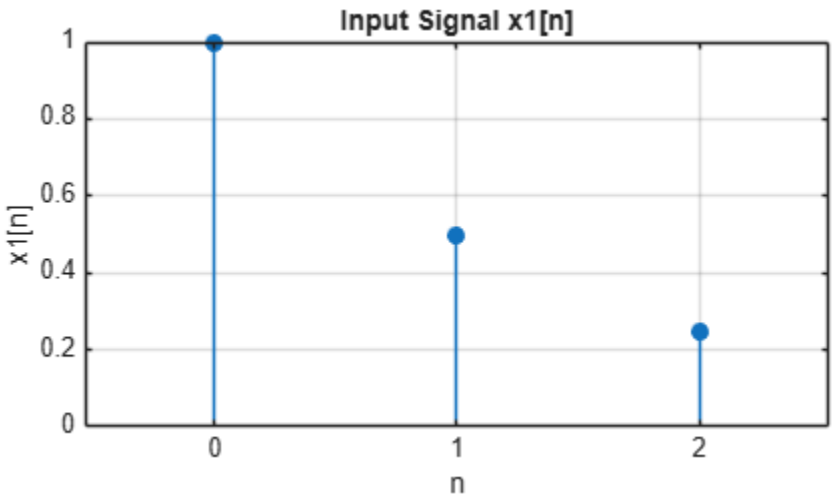
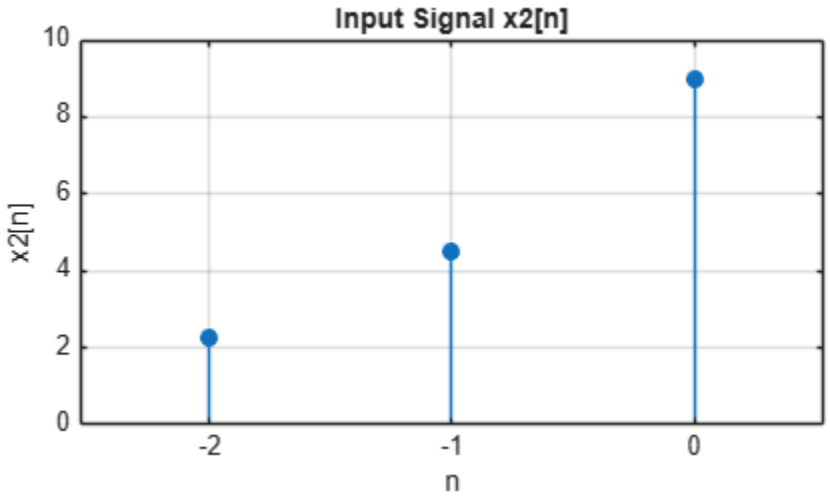
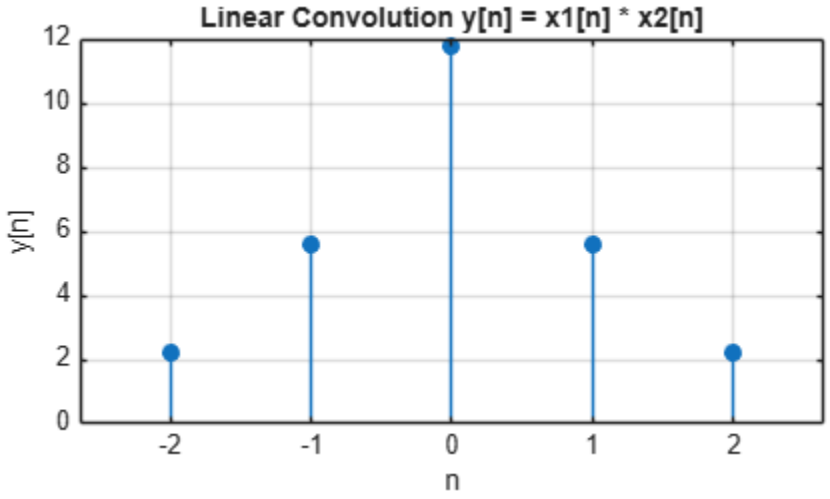
```
clc;
clear;
close all;

% Registration Number: 24103156019
% N1 = second last digit = 1
% N2 = last digit = 9

N1 = 1;
N2 = 9;

% Define input sequences
x1 = [N1, N1/2, N1/4];
x2 = [N2/4, N2/2, N2];

% Indexing of signals
n1 = 0:2;       % x1[n] indices: 0, 1, 2
n2 = -2:0;      % x2[n] indices: -2, -1, 0

% Perform convolution
y = conv(x1, x2);

% Indexing of convolution output
ny = (n1(1) + n2(1)) : (n1(end) + n2(end));

% Display sequences
disp('x1[n] = ');
disp(x1);

disp('x2[n] = ');
disp(x2);

disp('Convolution output y[n] = ');
disp(y);

disp('Indices of y[n] = ');
disp(ny);

% Plot x1[n]
figure;
stem(n1, x1, 'filled');
grid on;
xlabel('n');
ylabel('x1[n]');
title('Input Signal x1[n]');
xticks(0:2);

% Plot x2[n]
figure;
stem(n2, x2, 'filled');
grid on;
xlabel('n');
```

```
ylabel('x2[n]');
title('Input Signal x2[n]');
xticks(-2:0);

% Plot y[n]
figure;
stem(ny, y, 'filled');
grid on;
xlabel('n');
ylabel('y[n]');
title('Linear Convolution y[n] = x1[n] * x2[n]');
xticks(-2:2);
```

```
x1[n] =
    1.0000    0.5000    0.2500

x2[n] =
    2.2500    4.5000    9.0000

Convolution output y[n] =
    2.2500    5.6250   11.8125    5.6250    2.2500

Indices of y[n] =
    -2    -1     0     1     2
```







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